Software Security Analysis

Create Pseudocode and Flowchart for a Simple Program

Introduction

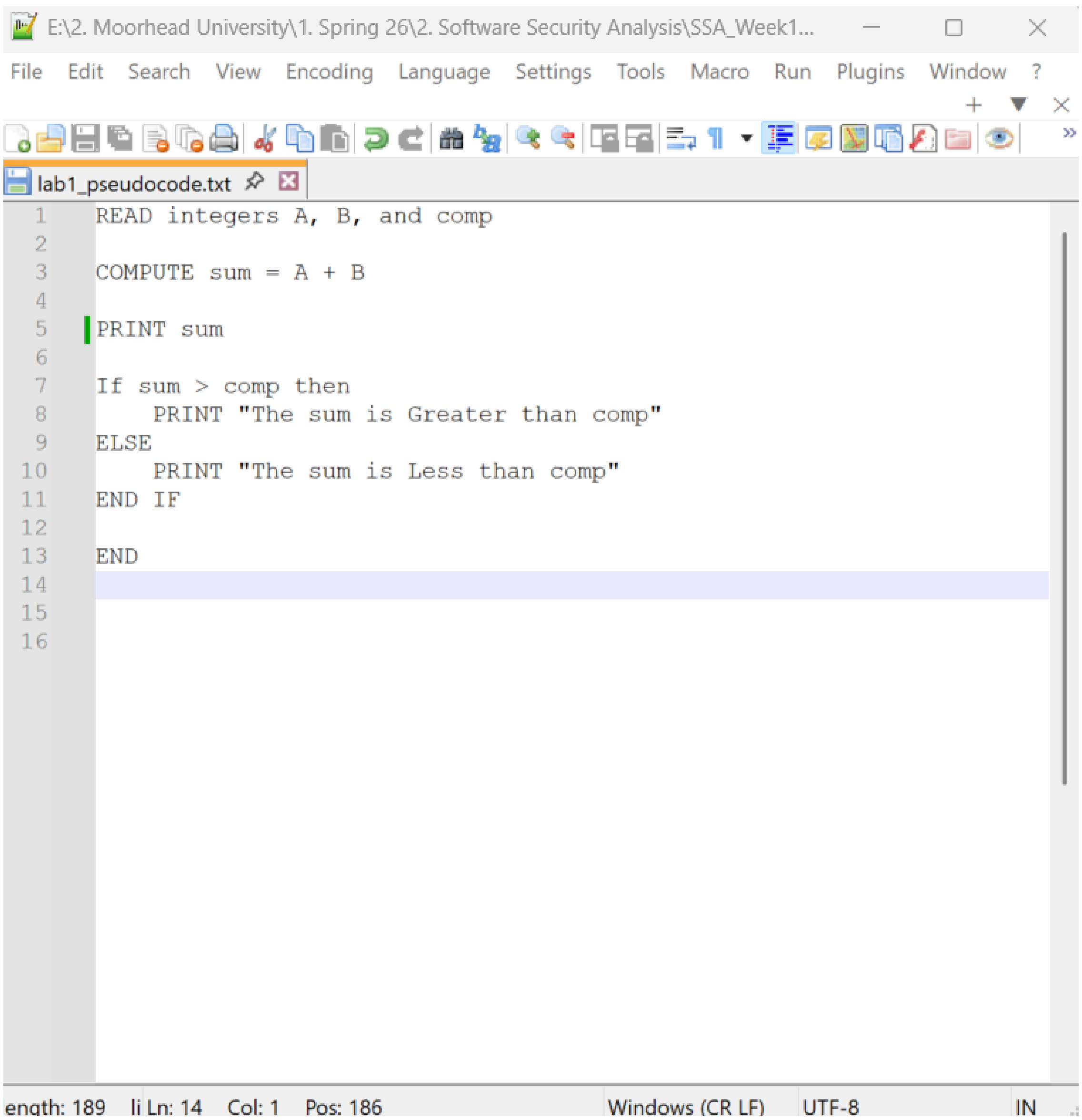
This lab is designed to work with Notepad++ and diagrams.net application to develop an understanding of fundamental programming concepts through pseudocode and flowchart. Before writing a program, it is important to plan and structure the flow of logic in a simple manner. Pseudocode is not a real programming code, but it models and may even look like a programming code (ITL Education Solutions Limited, 2011).

A *flowchart* is a pictorial representation of an algorithm in which the steps are drawn in the form of different shapes of boxes and the logical flow is indicated by interconnecting arrows (Domas, S., & Domas, C., 2024). This lab is designed to implement a simple program to read two integers, calculate their sum, and compare them with another integer called comp to check if the value is greater or less than the sum of two integers. The draw.io website was used to create a flowchart that visually represents the same logic using standard flowchart symbols.

1. Downloaded the Notepad++ using URL https://notepad-plus-plus.org/downloads/

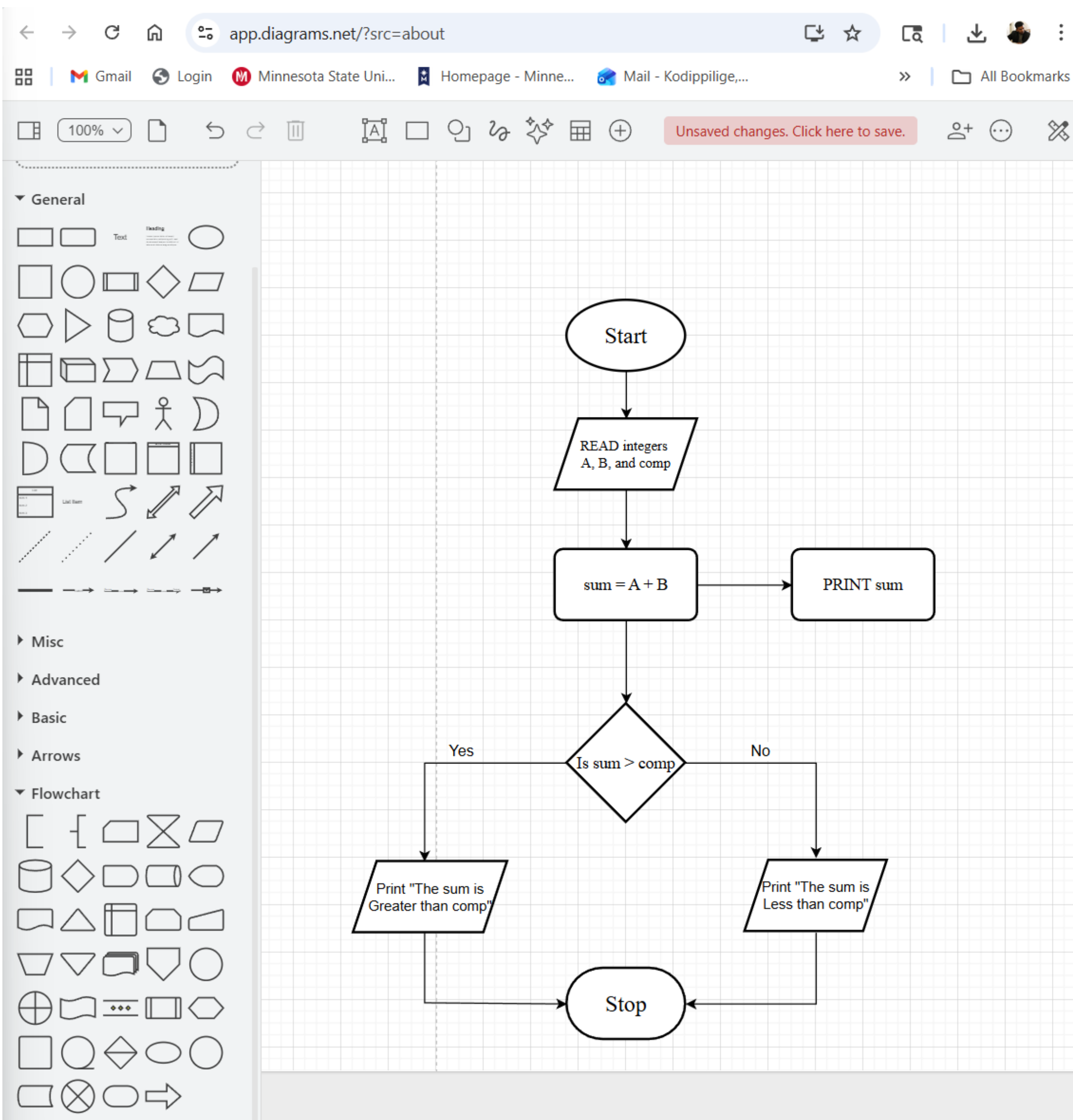
2. Installed the Notepad++ and started to write a pseudocode for simple program.

- Use A and B as two integers and comp also as a integer
- Add two integers together to get the “sum”
- Print the sum value
- Compare the value of “sum” and “comp”
- If sum is greater the comp Print “The sum is greater than comp”
- If not Print “The sum is less than comp”
- End the pseudocode



3. Created a flowchart using https://www.drawio.com/ website.

- Used Ellipse to Start the flowchart
- Used Parallelogram to Read integers
- Used Rectangle to process value of "sum"
- Used Rectangle to process Print of "sum"
- Used Diamond to take decision of if sum > comp
- Used Parallelogram to Print "The sum is Greater than comp" to left side of the Diamond if the sum > comp
- Used Parallelogram to Print "The sum is Less than comp" to right side of the Diamond if not sum > comp
- Used Terminator to Stop the flowchart.



Conclusion

This lab helps to cover the practical aspects of using Notepad++ and draw.io website for writing a simple pseudocode and creating a flowchart. Pseudocode provides a structured guideline for developing well-organized programs, while flowchart visually represents the program logic by showing the program's control flow and decision-making process in an easy-to-understand format.

Using pseudocode and flowchart helps minimize logical errors and improve clarity before implementing a program. We can understand difficult software security analysis using these techniques. Starting from easy program, we can gradually build a complex program using pseudocode and flowcharts as planning tools, reducing make major mistakes.

References

1. ITL Education Solutions Limited. (2011). *Introduction to computer science* (2nd ed.). Pearson Education India.
2. Domas, S., & Domas, C. (2024). *x86 software reverse-engineering, cracking, and counter-measures*. Wiley.
3. diagrams.net. (n.d.). *draw.io – Free online diagram software.* https://www.drawio.com/
4. Notepad++ Community. (n.d.). *Downloads*. https://notepad-plus-plus.org/downloads/
5. Lucid Software Inc. (n.d.). *What is a flowchart? Flowchart tutorial*. https://www.lucidchart.com/pages/what-is-a-flowchart-tutorial